

appropriate. Just make sure that you are consistent in your definitions throughout the pre-mortem exercise.

While conducting a pre-mortem workshop, participants must set and agree on rankings criteria upfront and then on the severity, occurrence, and detection level for each of the failure modes.

How to use the output of pre-mortem

The output of the pre-mortem analysis is only useful if you use it.

Each process step or system function would have one or more RPN values associated with it. The higher the RPN, the riskier the step is.

Before the pre-mortem exercise, the team must decide a threshold RPN value. For all the steps where RPN is above the threshold, risk mitigation and control plan become mandatory. For the ones below, risks may be addressed later as their priority would be lower.

Ideally, you should be addressing all the practical steps wherever RPN is non-zero. However, it is not always possible due to resource limitations.

One of the ways you can reduce RPN is by reducing the severity of the failure mode. Typically, reducing severity often needs functional changes in process steps or the solution itself. Additionally, the occurrence can be controlled by the addition of specific control measures such as a human in the loop or maker-checker mechanisms.

However, if it is not possible to reduce the severity or occurrence, you can contain the failures by implementing control systems. Control systems can help in the detection of causes of unwanted events.

Having risks quantified and visible enable you to have plans in place to act quickly and appropriately in case of failures and thus reduce the exposure to more failures or adverse consequences.

A common problem I have seen in this exercise is difficulty or failure to get to the root-cause of anticipated failure, and this is where SMEs should lean in. If you do not identify root-causes correctly, or do it poorly, your follow-up actions would not yield proper results.

Another problem I have seen is the lack of follow-up to ensure that recommended actions are executed, and the resulting RPN is lowered to an acceptable level. Doing effective follow-through is a project management function. It needs diligent execution to ensure that pre-mortem analysis reaches its logical conclusion.

Pre-mortem analysis workshops can be time-consuming at times. Due to high time demand, it may become challenging to get sufficient participation of SMEs. The key is to get the people who are knowledgeable and experienced about potential failures and their resolutions showing up at these workshops. SME attendance often needs management support, and facilitators need to ensure that this support is garnered.

Sector-specific considerations

In the pre-mortem analysis, severity-occurrence-detection (SOD) ratings range between one and ten. However, the weights assigned to each of the rating values are subjective. It is possible that in the same industry, two different companies could come up with slightly different ratings for the same failure mode.

If you want to avoid subjectivity and confusion, standardisation of the rating scale will be necessary. However, this would be only necessary when you are benchmarking two or more products from different vendors in the same industry. If this has to be used only for internal purposes, subjectivity will not matter much since relative weights will be still preserved within the risks and action items.

Nonetheless, when considering control or action plans for identified risks, sector-specific approaches could be (and should be) different.

Any failure risk can be controlled by reducing severity (S), lowering the chances of occurrence (O), or improving detection levels (D). If this were to be done in the banking sector while enhancing S and O ratings, D ratings might need additional focus for improvement.

Given the volume of transactions that the financial sector carries out every day, the severity of failure could be high due to widespread impact. But if the severity cannot be controlled

beyond a point, detecting it early to fix would be highly necessary.

For the healthcare sector, severity itself should be lower as the detection may likely result in fixing a problem but would not necessarily reverse the impact. For example, if AI prediction or solution results in an incorrect prognosis, early detection of this problem may result in stopping the activity per se. However, it will not be able to revert the issues caused by having this failure in the first place.

Similarly, in transportation scenario, especially for autonomous cars, detecting that a car's mechanism has failed as an after the fact is less useful since the accident would already have happened. Reducing severity and occurrence in those cases is a more acceptable course of action.

Severity and occurrence improvement are prevention-focused, whereas detection improvement is fixing (cure) focused. If your industry believes that prevention is better than cure, then work on reducing the severity and lowering the occurrence of failures. If your industry is comfortable with fixes after the fact, detection must be improved.

However, in my view, it is better to address all three factors and ensure that robust risk management is in place.

Conclusion

Pearl Zhu, in her book, *Digitizing Boardroom*, says, "Sense and deal with problems in their smallest state, before they grow bigger and become fatal."

Managing risks systematically and pragmatically is the key to handle AI risks. The problem with AI risks is that these are highly scalable and can quickly grow out of control due to the power of automation and the sheer capacity of AI to execute tasks.

Moreover, subjectivity in risk management is a myth. If you cannot quantify the risk, you cannot measure it. And if you cannot measure it, you cannot improve or control it. The systematic approach outlined here will help you quantify your risks, understand them better while maintaining the context of your use case.

You can develop and implement AI solutions responsibly. If you understand risks, understand them better and specific to your use case! **EFY**